

Sample Python Code to Generate a Global Map

```
# Generate a map of mean 2 m air temperature
# selected color map from cmocean colormaps for oceanography
cmap = cmocean.cm.thermal

# NumPy array for the color bar levels, here from -50 degC to +30 degC, in
increments of 2 degC
lev = np.arange(-50, 30, 2);

# map projection
proj = ccrs.Robinson(central_longitude=180)

# define figure and axes, figure size, and resolution (300 dpi)
fig = plt.figure(figsize=(9, 4.5), dpi=300)
ax = plt.axes(projection = proj)

# filled contour map of mean temperature (variable is m_t2m)
m_t2m.plot.contourf(
    x = 'lon',
    y = 'lat',
    ax=ax,
    transform=ccrs.PlateCarree(),
    levels=lev,
    extend='both',
    colors=cmap,
    add_colorbar=True,
    cbar_kwargs = {"label": "Temperature ( $^{\circ}$  C)"}))

# add coastlines
ax.add_feature(cfeature.BORDERS, linestyle=':')
ax.coastlines(
    resolution='110m') #Currently can be one of "110m", "50m", and "10m".

# add grid lines
gl = ax.gridlines(crs=ccrs.PlateCarree(),
                  draw_labels=True,
                  linewidth=1,
                  color='gray',
                  alpha=0.5,
                  linestyle='--')

# add title
ax.set_title("Mean 2m Temperature")

# save figure
fig.savefig('ERA5_mean_t2m.png', facecolor = 'white', transparent = False,
bbox_inches = 'tight')
```